

**A PROPOSED OFFERING OF
A VILLAGE RESIDENT INFORMATION & PROFILING SYSTEM
FOR DISIPLINA VILLAGE, UGONG**

A Project Proposal Presented to the
Faculty of Datamex College of Saint Adeline, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

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Maintenance

Documentation

INTRODUCTION

The Village Information System was developed to assist officials and staff in managing and organizing important data about the community. This includes resident profiles, household information. Instead of relying on manual record keeping, the system provides a faster and more accurate way of storing, updating, and retrieving information whenever needed. By using this system, the officials can improve its efficiency in serving residents and ensure that data is always well organized and secure.

Maintaining the system is an essential part of keeping it functional and reliable over time. Just like any other software, it needs regular attention to make sure it continues to run smoothly and securely. Through proper maintenance, the system can perform better, prevent technical problems, and stay updated with the latest technologies. This also helps in avoiding system errors, data loss, or security issues that could affect daily operations and services to the community.

The scope of maintenance for the Village Information System covers several important activities such as updating the software to the latest version, fixing bugs or system errors, applying security patches, backing up important data, and optimizing system performance. All of these efforts ensure that the system remains dependable, easy to use, and capable of supporting the long-term digital transformation of the barangay.

MAINTENANCE PLAN

The Maintenance Plan for the Village Information System focuses on keeping the system stable, secure, and responsive to the changing needs of the village. Since the system stores and manages important community data, it must always remain accessible and efficient for both administrators and staff. A well structured maintenance plan ensures that the system continues to perform smoothly, minimizes downtime, and adapts to future technological improvements.

Regular maintenance helps identify and solve technical issues before they affect daily operations. It also allows the development team to make updates, apply security measures, and improve overall performance. The plan is designed to be proactive meaning that the team doesn't wait for problems to happen, but rather takes steps to prevent them.

The Village Information System includes different types of maintenance, each serving a specific purpose:

Corrective Maintenance:

This involves fixing system bugs, software errors, or unexpected malfunctions reported by users. For example, if a form doesn't load properly or a button doesn't function as expected, corrective maintenance ensures it is resolved immediately.

Adaptive Maintenance:

As technologies evolve, the system needs to adapt to new platforms, operating systems, or software versions. Adaptive maintenance makes sure the system remains compatible with updated tools and can adjust to changes in the technological environment.

Perfective Maintenance:

This focuses on improving the overall performance and user experience of the system. It may include optimizing database speed, enhancing the user interface, or adding new features based on user feedback to make the system more efficient and user friendly.

Preventive Maintenance:

These are regular activities performed to prevent potential issues in the future. It includes scheduled data backups, code reviews, and system updates to ensure the software remains in good condition and continues to run without major problems.

Through this maintenance plan, the Village Information System can operate efficiently while ensuring the safety and reliability of all village data. Continuous monitoring, timely updates, and teamwork between the admin and developer play a big role in keeping the system dependable and sustainable for long-term use.

MAINTENANCE SCHEDULE

To keep the **Village Information System** running smoothly, a regular maintenance schedule is followed. This helps ensure that the system stays secure, updated, and performs well at all times. Having a clear schedule also allows the admin and support team to plan ahead and avoid unexpected issues that could interrupt daily operations.

Below is the maintenance schedule prepared for the system. Each task has its own frequency and assigned personnel to make sure every activity is completed properly and on time.

Task	Description	Frequency	Responsible Person	Status
Database backup	Regularly back up all data stored in the system to make sure important records are protected and can be restored if something goes wrong.	Weekly	Admin	Ongoing
Security Updates	Install the latest security patches and updates to protect the system from possible threats or unauthorized access.	Monthly	Development Team	Scheduled
Bug Fixes	Check for errors or issues reported by users and fix them to maintain a smooth and reliable system experience.	As Needed	Technical Support	Pending
System Performance Check	Review how the system performs, identify slow processes, and apply adjustments to keep it fast and stable.	Quarterly	IT Team	Not Started

ISSUE TRACKING & BUG REPORTS

Even with regular maintenance, some issues or minor errors may still occur while using the Village Information System. To handle these problems efficiently, an issue tracking process is followed. This allows the development and support teams to record, monitor, and resolve any technical concerns reported by users.

Keeping a record of issues helps the team identify common problems, understand their causes, and make improvements to avoid similar errors in the future. Each reported issue is given an ID and is tracked until it is completely fixed. This process ensures transparency and accountability in maintaining the system's performance and reliability.

Below is the sample record of issues encountered and their current status:

Issue ID	Description	Severity	Reported By	Date Reported	Status
BUG001	System sometimes freezes when saving resident data.	Critical	Admin	09/18/2025	In Progress
BUG002	Some resident records fail to update after editing	High	Staff	09/25/2025	Fixed
BUG003	Minor delay when loading the resident list page.	Low	Staff	10/10/2025	Fixed

BACKUP AND RECOVERY PLAN

The Village Information System contains essential records such as resident profiles, household data, and other community information. To protect these valuable files, a reliable backup and recovery plan is implemented. This ensures that even in cases of technical issues, hardware failure, or accidental deletion, all important data can be safely restored without major disruption.

Regular backups help keep the system secure and prevent the loss of information that could affect barangay operations. The plan also includes clear recovery steps to quickly bring the system back to normal if any issue occurs.

BACKUP PROCEDURES

The backup process for the Village Information System is simple but consistent. It focuses on keeping recent copies of the database.

Frequency of Backups

Backups are performed weekly. This schedule was chosen since most updates and data entries are completed by then. In addition, the admin performs a manual backup whenever there are important updates, new resident records, or changes to system settings.

Storage Locations

The backup files are stored in two secure places: one on an external hard drive connected to the main computer, and another in cloud storage such as Google Drive for easy recovery.

RECOVERY STEPS

If data loss or a system failure occurs, the following steps will be taken to restore the system and minimize downtime:

1. **Identify the issue** – Determine the reason for the data loss.
2. **Find the latest backup** – Locate the most recent working backup from either the external hard drive or the cloud storage.
3. **Restore the database** – Replace the corrupted or missing data files with the backup copy using the system's restore feature or database import function.
4. **Check the system** – Verify that all information and records have been restored correctly and that the system runs smoothly.
5. **Record the incident** – Document the recovery process and report it to the system developer or IT support to prevent the issue from happening again.

Technical Support Contact:

Development Team:

Email:

Phone:

PERFORMANCE MONITORING

Monitoring the performance of the Village Information System is an important part of maintenance. It helps ensure that the system continues to run smoothly, loads quickly, and responds properly to user actions. By keeping track of performance indicators, the admin and development team can identify potential issues early and take action before they affect daily operations.

Regular monitoring also helps maintain user satisfaction and prevents unnecessary downtime. The following key performance indicators (KPIs) are used to evaluate the overall health and stability of the system:

Metric	Description	Threshold	Monitoring Tool
Server Uptime	Measures the percentage of time the system is accessible and running properly.	99% or higher	Manual log checking and observation
Response Time	Time it takes for the system to load pages or display data after a user request.	Less than 3 seconds	System performance test
Error Rate	Tracks how often errors occur while using the system.	Below 1%	Error logs and user reports
Database Health	Checks if the database remains optimized and free of duplicate or unused records.	Clean database	SQL maintenance tools

SECURITY MEASURES

To protect the information stored in the Village Information System, several security measures are in place to prevent unauthorized access, data loss, and other potential threats. Since the system handles sensitive resident information, maintaining privacy and data integrity is a top priority.

Access Control - Only authorized users can log in to the system. There are two types of user roles:

- Admin – Has full control over all features, including editing, deleting, and managing user accounts.
- Staff – Has limited access and can only perform basic operations such as viewing and updating records.

Authentication - These measures help reduce the risk of others accessing the system without permission.

- All users are required to log in using a unique username and password.

Data Protection & Encryption

- Sensitive information such as resident details is stored securely in the database, protected from external access.

User Awareness - The system also promotes basic data security awareness among users. Staff are reminded to:

- Keep their login credentials private.
- Log out properly after use.
- Report any suspicious activity or unusual system behavior immediately to the admin.

DOCUMENTATION UPDATES

The documentation for the Village Information System is regularly reviewed and updated to reflect any improvements, fixes, or changes made to the system. Every time the system is modified such as when a new feature is added, a bug is fixed, or a security measure is improved the documentation is also updated to match those changes.

During the development and testing phases, several revisions were made to ensure clarity and accuracy. These include adding new sections like the backup and recovery plan, updating the maintenance schedule, and improving the explanation of performance monitoring and security policies.

Keeping the documentation updated helps both the admin and the developer maintain consistency and ensures that everyone involved understands the current state and capabilities of the system. This also makes future maintenance and system upgrades easier to implement.

CONCLUSION & RECOMMENDATIONS

The Village Information System was successfully developed and maintained to help simplify the process of managing resident and community records. Through proper maintenance, regular updates, and consistent monitoring, the system continues to perform efficiently and reliably in supporting day-to-day barangay operations.

The maintenance activities such as database backups, bug fixing, and performance monitoring play a big role in keeping the system stable and secure. These efforts also help prevent data loss and ensure that users can access information without interruptions. Overall, the maintenance plan has proven to be effective in extending the life and usability of the system.

Recommendations

To further improve the system and its long-term performance, the following recommendations are suggested:

1. **Implement an Automated Backup System** - Implement an automatic backup feature to lessen manual work and ensure data is always secured.
2. **Improve Security Features** - Adding features such as password recovery, login attempt limits, strengthens security and helps prevent unauthorized access.
3. **Enhance User Interface** – Improve the system's layout and navigation for better usability, especially for staff users.
4. **Add Notification Features** – Include alerts or reminders for upcoming maintenance schedules or system updates.
5. **Improve System Responsiveness** – Optimize database queries and clean up unused data to maintain smooth system performance as records continue to increase over time.

By following these recommendations, the Village Information System can continue to evolve into a more efficient, secure, and user friendly platform that benefits both administrators and staff members in managing community information.

